MUHAMMAD RIZWAN

Software Engineer | Full Stack & Mobile Development | Lahore, Pakistan
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PROFESSIONAL SUMMARY

Software Engineer with 4+ years of experience building secure, scalable backend systems and cross-platform mobile applications. Specialized in Node.js, Next.js, Python (FastAPI/Flask) and React Native, with hands-on delivery of enterprise authentication, search, and AI-integrated products serving 15,000+ active users.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL

Frameworks & Libraries: React Native, Next.js, Node.js, React.js, FastAPI, Flask

Databases & Search: MongoDB, MySQL, PostgreSQL, Elasticsearch

Developer Tools & AI: Git, JIRA, Docker, Postman, Google Colab

Core Technologies: Large Language Models (LLMs), AI Integrations, API Development

PROFESSIONAL EXPERIENCE

SeeBiz PVT LTD — Lahore, Pakistan

Software Engineer — July 2022 – Present

- Architected and deployed a centralized Enterprise Single Sign-On (SSO) ecosystem using Node.js and Next.js, mitigating authentication friction for over 15,000 active users across platform applications.
- Designed and implemented robust security layers, including cryptographic account lifecycle workflows and multi-factor authentication (MFA), reducing unauthorized access attempts by 99.9%.
- Boosted platform search efficiency by integrating an Elasticsearch cluster, reducing query latency by 40% for administrative data retrieval workflows.
- Transitioned legacy backend workflows into a serverless architecture, reducing cloud infrastructure costs by 30% and accelerating production deployment pipelines by 25%.
- Spearheaded core backend API development using Flask, FastAPI, and MongoDB for production mobile applications, cutting API endpoint response latency to under 200ms.

Software Engineer Intern — Jan 2022 – June 2022

- Collaborated with cross-functional engineering teams to model 10+ clean database schemas and scalable backends using MySQL and MongoDB, ensuring optimal data organization.
- Built JWT-based authorization layers and strict input validation schemas across 5+ microservices, strengthening security protocols and preventing unauthorized API access.
- Participated in agile sprints, frontend-backend integration, and debugging routines to maintain 99.9% uptime during feature deployments.

PROJECTS

Enterprise SSO System | Node.js, Next.js, Elasticsearch, Serverless, Swagger

- Constructed a secure identity management system with user profile dashboards, subscription modules, and account management workflows.
- Incorporated user activity audit trails, secure account deletion workflows, multi-factor authentication, and advanced search indices for administrative control.

SeeCard Platform | React Native, REST APIs, QR Code Infrastructure

- Built a dynamic digital business card mobile application supporting instant profile customization, photo integration, and real-time QR scanning workflows.
- Created smart contact synchronization enabling one-click saving to the device phonebook via dynamic QR codes, email, or web links.

SeeTask System | React Native, Real-time State, Multi-Tenant Architecture

- Launched a productivity and workspace application supporting multi-project assignment, team management, and secure user profiles with theme toggles.
- Formulated task lifecycles from ideation to completion with dynamic deadline trackers and low-latency real-time workspace notifications.

Turnfeat Platform | React Native, Node.js, PostgreSQL, OpenAI

- Conceptualized an AI-powered freelance marketplace, building high-availability database architectures for project posting and bidding pipelines.
- Merged collaborative team workspaces, secure milestones, automated escrow management, and contextual AI skill matching to reduce hiring friction.

PAKEZY Tourism Ecosystem | React Native, Python, FastAPI, MongoDB

- Devised a production-ready cross-platform travel ecosystem for regional tourism management, improving native rendering speeds and view hierarchies.
- Optimized location-based backend queries and map microservice responses, cutting geo-spatial query latency by 35% with high-frequency asset caching.

Utility Hub | React Native, Native File Modules

- Launched a client-side mobile utility application enabling offline Image-to-PDF compilation and lossless image compression.
- Implemented asynchronous device storage file mapping via native module bridges, decreasing runtime memory overhead during batch asset processing.

AI Predictive Sales Analytics | Python, BERT, TAPAS, Random Forest, Colab

- Programmed a data intelligence solution using Random Forest regression to forecast seasonal product demand and visualize complex variables.
- Authored an AI conversational retrieval engine leveraging fine-tuned BERT and TAPAS architectures for contextual product recommendations and semantic search.

EDUCATION

Gomal University — D.I. Khan, KPK

Bachelor of Science in Software Engineering — 2017 – 2021

Govt. College Boys Fatehpur — Layyah

FSc (Pre-Engineering) — 2015 – 2017